MAT 221 - CALCULUS I
Lesson 6 - 7: - Integrals of Certain Trigonometric Functions

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Integrals Involving Powers of Sine and Cosine

Guidelines for evaluating integrals involving powers of Sine and Cosine

1. When the power of the sine is odd and positive, save one sine factor and convert the remaining factors to cosines. Then, expand and integrate.

$$\int \sin^{2k+1} x \cos^n x \, dx = \int (\sin^2 x)^k \cos^n x \sin x \, dx$$
$$= \int (1 - \cos^2 x)^k \cos^n x \sin x \, dx$$

2. When the power of the cosine is odd and positive, save one cosine factor and convert the remaining factors to sines. Then, expand and integrate.

$$\int \sin^m x \cos^{2k+1} x \, dx = \int \sin^m x (\cos^2 x)^k \cos x \, dx$$
$$= \int \sin^m x (1 - \sin^2 x)^k \cos x \, dx$$

3. When the powers of both the sine and cosine are even and nonnegative, make repeated use of the identities

$$\sin^2 x = \frac{1-\cos 2x}{2} \quad \textit{and} \quad \cos^2 x = \frac{1+\cos 2x}{2}$$

to convert the integrand to odd powers of the cosine. Then proceed as in the second guideline.

Example 1

Calculate

(a) $\int \sin^3 x \cos^{\frac{4}{3}} x \, dx$ (b) $\int_0^{\frac{\pi}{2}} \sin^2 x \cos^4 x \, dx$

(c) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^3 x}{\sqrt{\sin x}} \, dx$ (d) $\int \cos^4 x \, dx$

Solution

(a) We write $\sin^3 x = (\sin^2 x)\sin x = (1-\cos^2 x)\sin x$. Then

$$\begin{aligned}\int \sin^3 x \cos^{\frac{4}{3}} x \, dx &= \int (1-\cos^2 x)\sin x \cos^{\frac{4}{3}} x \, dx \\ &= \int \cos^{\frac{4}{3}} x \sin x \, dx - \int \cos^{\frac{10}{3}} x \sin x \, dx \\ &= -\frac{3}{7}\cos^{\frac{7}{3}} x + \frac{3}{13}\cos^{\frac{13}{3}} x + C\end{aligned}$$

(b) Using the identity $2\sin x\cos x = \sin 2x$ and $\cos^2 x = \frac{1+\cos 2x}{2}$.

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \sin^2 x\cos^4 x\,dx &= \int_0^{\frac{\pi}{2}} (\sin x\cos x)^2\cos^2 x\,dx \\
&= \int_0^{\frac{\pi}{2}} \left(\frac{1}{2}\sin 2x\right)^2 \left(\frac{1+\cos 2x}{2}\right)\,dx \\
&= \frac{1}{8}\int_0^{\frac{\pi}{2}} \sin^2 2x\,dx + \frac{1}{8}\int_0^{\frac{\pi}{2}} \sin^2 2x\cos 2x\,dx \\
&= \frac{1}{16}\int_0^{\frac{\pi}{2}} (1-\cos 4x)\,dx + \frac{1}{8}\cdot\frac{1}{3}\cdot\frac{1}{2}\sin^3 2x\Big|_0^{\frac{\pi}{2}} \\
&= \frac{1}{16}\left[x - \frac{\sin 4x}{4}\right]_0^{\frac{\pi}{2}} + \frac{1}{48}(\sin^3\pi - \sin^3 0) \\
&= \frac{\pi}{32}
\end{aligned}$$

(c)-(d) in class.

Integrals Involving Powers of Secant and Tangent

Guidelines for evaluating integrals involving powers of Secant and Tangent

1. When the power of the secant is even and positive, save a secant-squared factor and convert the remaining factors to tangents. Then, expand and integrate.

$$\begin{aligned}\int \sec^{2k} x \tan^n x \, dx &= \int (\sec^2 x)^{k-1} \tan^n x \sec^2 x \, dx \\ &= \int (1 + \tan^2 x)^{k-1} \tan^n x \sec^2 x \, dx\end{aligned}$$

2. When the power of the tangent is odd and positive, save one secant-tangent factor and convert the remaining factors to secants. Then, expand and integrate.

$$\begin{aligned}\int \sec^m x \tan^{2k+1} x \, dx &= \int \sec^{m-1} x (\tan^2 x)^k \sec x \tan x \, dx \\ &= \int \sec^{m-1} x (\sec^2 x - 1)^k \sec x \tan x \, dx\end{aligned}$$

3. When there are no secant factors and the power of the tangent is even and positive, convert a tangent-squared factor to a secant-squared factor, then expand and repeat if necessary.

$$\begin{aligned}\int \tan^n x\,dx &= \int \tan^{n-2} x(\tan^2 x)\,dx \\ &= \int \tan^{n-2} x(\sec^2 x - 1)\,dx\end{aligned}$$

4. When the integral is of the form

$$\int \sec^m x\,dx$$

where m is odd and positive, use integration by parts.

5. When none of the first four guidelines applies, try converting to sines and cosines

Example 2

Calculate

(a) $\int \tan^2 x\, dx$ (b) $\int_0^{\frac{\pi}{3}} \tan^7 x \sec^4 x\, dx$ (c) $\int_0^{\frac{\pi}{4}} \tan^4 x\, dx$

(d) $\int \tan^2 x \sec x\, dx$ (e) $\int \frac{\tan^3 x}{\sqrt{\sec x}}\, dx$ (f) $\int \tan^3 3x \sec^4 3x\, dx$

(g) $\int \frac{\sec x}{\tan^2 x}\, dx$ (h) $\int \sec 4x\, dx$

Solutions

(a) $\int \tan^2 x\, dx = \int (\sec^2 x - 1)\, dx = \tan x - x + C$

(b)

$$\int_0^{\frac{\pi}{3}} \tan^7 x \sec^4 x\, dx = \int_0^{\frac{\pi}{3}} \tan^7 x(1 + \tan^2 x) \sec^2 x\, dx$$

$$= \int_0^{\frac{\pi}{3}} \tan^7 x \sec^2 x\, dx + \int_0^{\frac{\pi}{3}} \tan^9 x \sec^2 x\, dx$$

$$= \left(\frac{\tan^8 x}{8} + \frac{\tan^{10} x}{10} \right)_0^{\frac{\pi}{3}} = \frac{1377}{40}.$$

(c)-(h) In class

Products of Sines and Cosines

Theorem 1

$$\sin mx \sin nx = \frac{1}{2}(\cos[(m-n)x] - \cos[(m+n)x])$$

$$\sin mx \cos nx = \frac{1}{2}(\sin[(m-n)x] + \sin[(m+n)x])$$

$$\cos mx \cos nx = \frac{1}{2}(\cos[(m-n)x] + \cos[(m+n)x])$$

Example 3

Find

(a) $\int \sin 5x \cos 4x \, dx$ (b) $\int \cos 6x \cos 4x \, dx$ (c) $\int \sin 5x \sin x \, dx$

solution

(a)

$$\int \sin 5x \cos 4x \, dx = \frac{1}{2}\int (\sin x + \sin 9x) dx = -\frac{1}{2}\cos x - \frac{1}{18}\cos 9x + C$$

(b)-(c) In class